REMARKS ON THE SPORADIC RATIONAL TETRAHEDRA THAT MIGHT TILE SPACE

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Abstract. Chentouf and Sun showed that at most 40 of the 59 sporadic rational tetrahedra of Kedlaya, Kolpakov, Poonen and Rubinstein can tile $\mathbb{R}^3$, and they left these 40 undecided. We make six remarks.

(1) The printed list contains 42 tuples, two of which are excluded in the body of their paper.
(2) Two members of the list are classical space-fillers, namely Sommerville's tetrahedra No. 2 and No. 3.
(3) None of the remaining 38 tetrahedra admits a face-to-face tiling. For 12 of them this is new.
(4) In any tiling by one of the 38, faults are forced along every copy of certain edges.
(5) Three members are unions of two mirror copies of a fourth member, #36, which is the corner of a golden box. A golden box splits into four copies of #36 and one copy of #7.
(6) Constructive searches for non-face-to-face tilings find nothing for the 38. The searches cover clusters of up to 7 copies, rods (infinite prisms filled by a screw orbit of a motif of up to 3 copies) and slabs (layers with a motif of up to 6 copies). They do find tilings for the controls.

We also explain why purely local arguments cannot settle the icosahedral members of the list, and we record the local structure of #36 at its only bad edge.

1. Setting

A tetrahedron is *rational* if all of its dihedral angles are rational multiples of π. Kedlaya, Kolpakov, Poonen and Rubinstein [5] classified such tetrahedra into two one-parameter families and 59 sporadic examples. Since rational tetrahedra have Dehn invariant zero, Debrunner's theorem [3] gives no obstruction to tiling. Chentouf and Sun [2] gave three necessary criteria:

- their Proposition 2.3, for tilings that are not face-to-face;
- their Proposition 2.7, for face-to-face tilings;
- a length-weighted linear program, their Corollary 2.13.

They proved that the sporadic tetrahedra that tile lie in a set $\mathscr{A}$ of 40 tetrahedra (their Theorem 1.2). Whether the members of $\mathscr{A}$ tile was left open.

We use their conventions. Vertices are labelled $1, 2, 3, 4$, α_{ij} is the dihedral angle along edge ij, and angles are listed in the order $(\alpha_{12}, \alpha_{34}, \alpha_{13}, \alpha_{24}, \alpha_{14}, \alpha_{23})$ as multiples of π. A tiling is *face-to-face* (f2f) if no edge of a tile meets the relative interior of a face of another tile [2, Def. 2.2]. Tilings may use mirror images. All computations are available at [7] (directory `research/p2`).

2. The list

We number the 42 tuples printed in [2, Thm. 1.2] as $\#1, \ldots, \#42$ in reading order. Tuples #34 and #35,

$$\left(\tfrac{3}{20}, \tfrac{11}{20}, \tfrac{11}{20}, \tfrac{1}{4}, \tfrac{1}{3}, \tfrac{2}{3}\right) \quad \text{and}$$
$$\left(\tfrac{11}{60}, \tfrac{31}{60}, \tfrac{11}{20}, \tfrac{1}{4}, \tfrac{3}{10}, \tfrac{7}{10}\right),$$

are exactly the two tetrahedra that [2, §3.1, eq. (5)] excludes by the linear program. So the printed set has 42 elements, and the intended $\mathscr{A}$ consists of the other 40.

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We checked every tuple as follows:

- the Gram matrix $\big(-\cos\alpha\big)$ is singular to within 10^{-40} and positive semidefinite, with a positive kernel vector;
- the dihedral angles recomputed from the reconstructed vertices agree with the data to within 10^{-60}.

This guards against transcription errors. Our implementation of the linear program of [2, Cor. 2.13] excludes precisely #34 and #35 and is feasible for the other 40.

3. Two classical tilers in $\mathscr{A}$

Proposition 1. *Tetrahedron* $\#1 = (\frac{1}{4}, \frac{1}{3}, \frac{1}{3}, \frac{1}{4}, \frac{1}{2}, \frac{2}{3})$ *is Sommerville's No. 3. Tetrahedron* $\#3 = (\frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, \frac{1}{2})$ *is Sommerville's No. 2, that is, the Coxeter simplex* $[4, 3^{1,1}]$. *Both tile* $\mathbb{R}^3$.

Proof. In the notation of Table 1 of [1], Sommerville No. 3 has angles $2\pi/n_{ij}$ with

$$(n_{12}, n_{13}, n_{14}, n_{23}, n_{24}, n_{34}) = (3, 6, 6, 8, 8, 4).$$

Relabelling its vertices by $1 \mapsto 3$, $3 \mapsto 1$ gives #1. Sommerville No. 2 has $(4, 4, 4, 6, 6, 8)$, and the relabelling $1 \mapsto 3$, $2 \mapsto 4$, $3 \mapsto 1$, $4 \mapsto 2$ gives #3. A tetrahedron is determined up to similarity by its ordered dihedral angles.

Independently, we verified tilings with the Poincaré polyhedron theorem. For #3, the reflections in all four faces satisfy the edge conditions: every dihedral angle is π/m, so this is the Coxeter group $[4, 3^{1,1}]$. For #1, the following pairing satisfies all edge-cycle conditions exactly:

- the faces opposite vertices $1, 2, 3$ are mirrors;
- the face opposite vertex 4 is paired with itself by the half-turn about its axis of symmetry.

In all, our search finds 6 isohedral face pairings for #1 and 4 for #3. □

Hence at most 38 members of $\mathscr{A}$ are genuinely undecided.

4. No face-to-face tilings

Lemma 2. *In a face-to-face tiling by tetrahedra, any two tiles meet in a common face, edge or vertex, or not at all. In particular, the tiles containing a point in the relative interior of an edge E of a tile all have E as an edge, and consecutive tiles around E share a whole face.*

Proof. If two coplanar faces overlap but are not equal, the boundary of one meets the interior of the other [2, Lemma 2.6]. This is excluded, so overlapping coplanar faces coincide.

Let p lie in the relative interior of an edge E of a tile A. Let F be a face of A containing E. The tile across F near p has a face overlapping F, hence equal to F, so it contains E as an edge. Repeating this around E shows that every tile containing p has E as an edge. The remaining cases follow in the same way. □

Definition 3. Fix an edge length ℓ. A *wedge* is a tile together with one of its edges ab of length ℓ, placed on a fixed segment PQ (with $a \mapsto P$, $b \mapsto Q$ or the reverse), and an ordering $(F_{\mathrm{in}}, F_{\mathrm{out}})$ of its two faces through ab. The angle of the wedge is α_{ab}.

A face through PQ is described by the pair $(|P\,x|, |Q\,x|)$, where x is its third vertex. The *star graph* of ℓ has these pairs as vertices and the wedges as arcs from F_{in} to F_{out}. An *edge star* of an edge e of T is a closed walk in the star graph of total angle 2π that uses a wedge of T on e.

Mirror images are taken into account automatically, because only metric data enter.

Proposition 4. *If T admits a face-to-face tiling, then every edge of T has an edge star.*

Proof. By Lemma 2, the tiles around a copy of an edge e form a cyclic sequence of wedges on that edge. Consecutive wedges share a whole face, so the face data match, and the angles sum to 2π. □

Proposition 2.7 of [2] is the special case of Proposition 4 for tetrahedra whose edge lengths are pairwise distinct or parallelogram-like. In that case the star graph forces repeated reflections.

Theorem 5. *None of the* 38 *tetrahedra in* $\mathscr{A} \setminus \{\#1, \#3\}$ *admits a face-to-face tiling. For* $\#2, 5, 7, 12, 19, 20, 22, 23, 25, 27, 36, 42$*, this does not follow from* [2, Prop. 2.7].

Proof. For each of the 38 tetrahedra, at least one edge has no edge star; Table 1 lists these edges. Checking this is a finite search in a graph with at most 12 vertices, carried out over $\mathbb{Q}$ for the angles. As a control, every edge of #1 and #3 has an edge star.

Edge lengths are computed to 60 digits. Distinct length classes differ by at least 0.0093, and equal lengths agree to 10^{-59}. Wrongly identifying two lengths could only add arcs to the star graph, so the non-existence statements are robust under this identification.

For the last claim, the 12 listed tetrahedra are exactly those in which the edge lengths are neither pairwise distinct nor parallelogram-like, or in which every dihedral angle divides 2π. For these, Proposition 2.7 of [2] does not apply. □

Remark 6. None of these tetrahedra can be excluded by Proposition 2.3 of [2], because each of the 40 has several dihedral π-combinations. So every potential tiling is necessarily non-face-to-face.

5. Forced faults

Call an edge of T *bad* if it has no edge star. A *metric chain* is a path in the star graph through a wedge of T on that edge; its angle is the sum of the wedge angles. For all 38 tetrahedra, no bad edge has a metric chain of total angle exactly π.

Take a tiling and a generic point p on a copy of a bad edge. The tiles around p split into maximal metric chains. Between two consecutive chains there is a *fault half-plane*, in which the faces of the two adjacent tiles are different triangles. Alternatively, p lies in the relative interior of a face of another tile. Hence:

Proposition 7. *In any tiling by one of the* 38 *tetrahedra, every point of every copy of a bad edge lies on a fault half-plane, or in the interior of a face of another tile. In particular, faults occur along a set of positive length in every tile.*

Clusters. A natural way to produce faults is to glue copies of T face-to-face into a convex polytope Q that tiles. Neighbouring copies of Q may then triangulate their common faces differently. Interior edges of Q need edge stars, and edges inside faces of Q need metric chains of angle π. Therefore every bad edge lies on an edge of Q, with Q's dihedral angle there equal to a metric-chain angle.

We enumerated all clusters of at most 7 copies glued face-to-face (mirror images allowed), up to congruence. For each convex cluster we tested the Poincaré conditions over all pairings of its polygonal faces by isometries, including translations, half-turns, reflections and point reflections:

- for #1 and #3 the method finds many tiling clusters;
- for the other 38 there are only 0–24 convex clusters (last column of Table 1), and none tiles.

6. Icosahedral members

Many members of $\mathscr{A}$ are icosahedral. Take, for example, $\#19 = (\frac{1}{5}, \frac{1}{5}, \frac{2}{5}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3})$ and $\#25 = (\frac{1}{5}, \frac{1}{3}, \frac{1}{3}, \frac{1}{5}, \frac{1}{2}, \frac{4}{5})$.

- Every dihedral angle lies in $\{\frac{\pi}{5}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{5}\}$.
- Every vertex figure is a union of 1 to 14 copies of the Schwarz triangle $(2, 3, 5)$, which tiles the sphere. Vertex 4 of #19 and vertices $1, 4$ of #25 *are* this triangle.
- The golden ratio appears among the edge lengths: $|24|/|23| = \varphi$ for both, and also $|12|/|23| = \varphi$ for #25.

In both, the only bad edge is 23, of unique length.

Every local angular condition is satisfiable for these tetrahedra:

- the edge combinations around every edge;
- spherical vertex stars;
- planar vertex stars in fault planes, with the compatibility condition that $\pi - \alpha_g - \alpha_{g'}$ is a non-negative integer combination of dihedral angles across each interior edge;
- the length-weighted linear program, which has no forced combination.

An obstruction, if one exists, must therefore be global and metric. The distinct edge lengths are linearly independent over $\mathbb{Q}$ numerically (PSLQ finds no relation with coefficients up to 10^6). This suggests counting edge lengths along lines that carry faults.

7. The golden box

Tetrahedra #25, #27 and #42 have a mirror symmetry exchanging vertices 1 and 4. In each case the mirror cuts the tetrahedron into two congruent halves, and every half is

$$\#36 = (\tfrac{1}{5}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{2}{5}, \tfrac{1}{2}).$$

So if #36 does not tile, then neither do #25, #27, #42.

The three dihedral angles at vertex 3 of #36 are right angles. The edges $34, 31, 32$ are therefore mutually perpendicular, with lengths in the ratio $1 : \varphi : \varphi^2$, where φ is the golden ratio. Hence #36 is the trirectangular corner of the box $[0,1] \times [0,\varphi] \times [0,\varphi^2]$.

Proposition 8. *The golden box is the union of four copies of* #36 *(up to mirror image), cut off at alternate vertices, and one copy of* $\#7 = (\tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{3}, \tfrac{1}{3}, \tfrac{3}{5}, \tfrac{3}{5})$*. In particular,* #36 *and* #7 *together tile* $\mathbb{R}^3$ *periodically.*

This was verified to 50 digits. #7 is not a union of copies of #36: its faces are acute, while every cluster of copies of #36 exposes right triangles. Whether either tetrahedron tiles alone is open. A search over all clusters of up to 6 copies finds no other dissection relation within $\mathscr{A}$.

8. Searches for non-face-to-face tilings

Every tiling by one of the 38 is non-face-to-face. We searched for the most natural constructions of such tilings. In each case, controls with known tilings are found.

Non-convex clusters. We enumerated all clusters of at most 5 copies glued face-to-face, convex or not. We kept those in which every bad edge of every copy lies on an edge of the cluster. For each we tested the Poincaré conditions over all pairings of the polygonal faces, computing the dihedral angles exactly, and checked every hit by covering random points. For #1 and #3 many tilings are found. For the 38 there are none.

Rods. Let g be an isometry that glues a motif M (a cluster of copies) to $g(M)$ along a whole face. Suppose the orbit $\{g^k M\}$ fills an infinite convex prism. Then such prisms, placed along any tiling of the plane by the cross-section and slid along their axes at will, tile $\mathbb{R}^3$, generically not face-to-face. The orbit fills a prism if and only if

$$\text{area of the projection along the screw axis} \times \text{axial step} = (\text{copies per step}) \cdot \mathrm{vol}(T).$$

For an improper g we use g^2. We tested all such g for motifs of 1, 2 and 3 copies.

- A Hill-type tetrahedron cut from a triangular prism along a helix gives rods, as it should.
- #1 gives 16 rods with 2-copy motifs.
- None of the 38 gives a rod.

Slabs. Let a motif of $k \leq 6$ copies, with a lattice of translations $\langle t_1, t_2 \rangle$, fill a slab between two parallel planes. Then slabs stacked with arbitrary offsets tile $\mathbb{R}^3$. Candidate translations glue the motif to a translate along a pair of opposite free faces. The criterion is

$$k \cdot \mathrm{vol}(T) = \mathrm{thickness} \cdot |t_1 \times t_2|,$$

together with non-overlap. The cube orthoscheme gives slabs of cubes with $k = 6$. None of the 42 tetrahedra gives a slab with $k \leq 6$, including #1 and #3. So this search is limited by the size of the motif.

9. The local structure of #36

Place #36 in the golden box inside the standard icosahedral frame. Then every edge lies on a rotation axis of the icosahedral group:

- the legs $31, 32, 34$ lie on 2-fold axes, with dihedral angle $\pi/2$;
- edge 24 lies on a 3-fold axis, with $\pi/3$;
- edges 12 and 14 lie on 5-fold axes, with $\pi/5$ and $2\pi/5$.

The six edge lengths are distinct. The only parallel pair is 12 and 14, with $|12| = \varphi\,|14|$.

The only bad edge is 14. Its two faces, 134 (a right triangle with legs 1 and φ) and the slanted face 124, are not congruent. So consecutive wedges in a face-to-face cycle alternate between the two face types, and the cycle length must be even. But $2\pi/(2\pi/5) = 5$ is odd.

Measure angles in units of $\pi/30$. Around a point of a copy of edge 14, the other tiles must contribute 48 in total, using 6 (edge 12), 10 (edge 24), 12 (edge 14), 15 (a leg) and 30 (a face). There are exactly 11 ways. The pure five-fold cycle forces a mismatched half-plane.

Dehn-type invariants vanish identically for rational tetrahedra: they take values in $\mathbb{R}\otimes\mathbb{Q}/\mathbb{Z} = 0$. Counting lengths along the 5-fold lines involves chords of faces of arbitrary length. A proof that #36 does not tile would therefore have to control the global arrangement. One possible route is to show that the tiles use only the 120 icosahedral orientations unless a whole fault plane exists.

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#	$(\alpha_{12}, \alpha_{34}, \alpha_{13}, \alpha_{24}, \alpha_{14}, \alpha_{23})/\pi$	edges w/o star	status	clusters
1	(1/4, 1/3, 1/3, 1/4, 1/2, 2/3)	–	tiles (Sommerville No. 3)	31
2	(5/24, 3/8, 1/3, 1/4, 13/24, 5/8)	12, 34, 14, 23	no f2f (new)	1
3	(1/4, 1/2, 1/2, 1/3, 1/3, 1/2)	–	tiles (Sommerville No. 2)	28
4	(7/24, 11/24, 13/24, 7/24, 1/3, 1/2)	12, 34, 13, 24	no f2f	3
5	(1/5, 1/5, 1/3, 1/5, 2/3, 2/3)	14, 23	no f2f (new)	0
6	(1/5, 1/5, 4/15, 4/15, 3/5, 11/15)	13, 24, 14, 23	no f2f	0
7	(1/5, 1/5, 1/3, 1/3, 3/5, 3/5)	14, 23	no f2f (new)	0
8	(1/3, 1/3, 3/5, 1/3, 2/5, 2/5)	13, 14, 23	no f2f	3
9	(4/15, 8/15, 1/3, 1/3, 7/15, 7/15)	12, 34, 14, 23	no f2f	2
10	(1/7, 3/7, 1/3, 1/3, 4/7, 4/7)	34, 14, 23	no f2f	0
11	(2/7, 2/7, 1/3, 1/3, 3/7, 5/7)	12, 34, 14, 23	no f2f	1
12	(1/5, 2/5, 1/2, 1/3, 1/3, 2/3)	34, 23	no f2f (new)	4
13	(2/15, 7/15, 1/2, 1/3, 2/5, 3/5)	12, 34, 14, 23	no f2f	2
14	(2/15, 7/15, 31/60, 19/60, 5/12, 7/12)	12, 34, 13, 24, 14, 23	no f2f	1
15	(1/5, 2/5, 7/12, 1/4, 5/12, 7/12)	34, 13, 14, 23	no f2f	1
16	(13/60, 23/60, 7/12, 1/4, 2/5, 3/5)	12, 34, 13, 14, 23	no f2f	1
17	(1/5, 3/5, 1/3, 1/3, 1/2, 1/2)	34	no f2f	4
18	(3/10, 7/10, 1/3, 1/3, 2/5, 2/5)	12, 34, 14, 23	no f2f	2
19	(1/5, 1/5, 2/5, 1/3, 1/2, 2/3)	23	no f2f (new)	5
20	(1/6, 7/30, 11/30, 11/30, 1/2, 2/3)	34, 13, 24, 23	no f2f (new)	5
21	(1/5, 1/5, 9/20, 17/60, 11/20, 37/60)	13, 24, 14, 23	no f2f	0
22	(1/5, 1/3, 1/2, 1/3, 2/5, 3/5)	14	no f2f (new)	6
23	(1/6, 11/30, 1/2, 1/3, 13/30, 17/30)	34, 14, 23	no f2f (new)	6
24	(1/6, 11/30, 29/60, 7/20, 5/12, 7/12)	34, 13, 24, 14, 23	no f2f	2
25	(1/5, 1/3, 1/3, 1/5, 1/2, 4/5)	23	no f2f (new)	3
26	(7/60, 5/12, 1/3, 1/5, 7/12, 43/60)	12, 34, 14, 23	no f2f	0
27	(1/5, 2/5, 2/5, 1/5, 1/2, 2/3)	34, 13	no f2f (new)	9
28	(1/5, 2/5, 1/3, 1/3, 1/2, 3/5)	34, 23	no f2f	3
29	(7/30, 13/30, 3/10, 3/10, 1/2, 3/5)	12, 34, 13, 24, 23	no f2f	3
30	(1/4, 7/20, 1/3, 1/3, 9/20, 13/20)	34, 14, 23	no f2f	2
31	(1/5, 1/2, 3/5, 1/5, 1/3, 2/3)	13, 23	no f2f	1
32	(1/5, 1/2, 17/30, 7/30, 3/10, 7/10)	13, 24, 14, 23	no f2f	1
33	(3/20, 11/20, 17/30, 7/30, 7/20, 13/20)	12, 34, 13, 24, 14, 23	no f2f	1
36	(1/5, 1/2, 1/2, 1/3, 2/5, 1/2)	14	no f2f (new)	24
37	(1/5, 1/2, 7/15, 11/30, 11/30, 8/15)	13, 24, 14, 23	no f2f	2
38	(4/15, 13/30, 17/30, 4/15, 2/5, 1/2)	12, 34, 13, 24, 14	no f2f	3
39	(4/15, 13/30, 3/5, 3/10, 11/30, 7/15)	12, 34, 13, 24, 14, 23	no f2f	3
40	(4/15, 17/30, 2/5, 3/10, 11/30, 8/15)	12, 34, 13, 24, 14, 23	no f2f	1
41	(3/10, 2/5, 3/5, 3/10, 1/3, 1/2)	12, 34, 13, 24	no f2f	3
42	(1/3, 2/5, 2/5, 1/3, 1/2, 2/5)	34, 13	no f2f (new)	10

TABLE 1. The 40 members of $\mathscr{A}$, numbered by position in the printed list of [2] (#34 and #35 omitted). "No f2f" means that no face-to-face tiling exists (Theorem 5); "new" means it is not covered by [2, Prop. 2.7]. "Clusters" is the number of convex face-to-face clusters of at most 7 copies, up to congruence.